

Identification (识别)

Specification (设定)

Estimation (估计)

Inference (推断)

参数是存在的、固定的、未知的

参数的识别: 参数之间必须有独立作用的空间 (α与β不能乘在一起)

设定:

$$\begin{cases} G = \alpha L + bC & \checkmark \\ G = L^{\alpha} C^{\beta} & \checkmark \\ G = \alpha^L \cdot \beta^C & \times \end{cases}$$

估计:

大数定律

$X_1, X_2, \dots, X_n$ be independent and identical distributed (i.i.d.),

given $E(X_i) < \infty$, then:

$$\hat{\mu} = \frac{\sum_{i=1}^n X_i}{n} \rightarrow \mu \text{ as } n \rightarrow \infty$$

中心极限定理 (CLT)

Let $X_1, X_2, \dots, X_n$ be independent and identical distributed:

with $E(X_i) = \mu < \infty$ and $\text{var}(X_i) = \sigma^2 < \infty$, then:

$$\sqrt{n}(\hat{\mu} - \mu) \rightarrow^D N(0, \sigma^2)$$

or:

$$\frac{\hat{\mu} - \mu}{\sqrt{\frac{\sigma^2}{n}}} \rightarrow N(0, 1)$$

↓
统计量: 相当于从标准正态里抽样一次, 一组样本只能估计一个 $\hat{\mu}$

不离谱不下结论

假设检验

本质: 先认为假设正确, 再按假设值计算, 代入统计量公式中; 计算结果不能离谱

Two type of error: ① 以真为假: $p = 1 - \alpha$ 显著性水平; ①应尽力避免, 概率为 α .

② 以假为真:

置信区间

最小二乘法

$$\min \sum_{i=1}^n (x_i - \alpha - \beta_1 C_i - \beta_2 L_i)^2$$

ARMA

Stochastic process

is a sequence of random variable.

$$\{x_t\}_{t=1}^T$$

data $\xrightleftharpoons[\text{decide}]{\text{inflect}}$ model

新信息与人决定 “有限注意理论”

Random Walk Hypothesis

$$P_t = \mu + P_{t-1} + \varepsilon_t$$

information {
 ① 简单有效信息 (K线) → 弱有效市场
 ② 公开信息
 ③ 内部消息
 ④ → 完全有效市场

① μ is a drift to show the stable increase of stock price

② ε is the idiosyncratic shock that can have persistent influence effects on the price. source of volatility

③ The property of P_t is determined by the properties of ε_t .

④ Use several ways to show this process is not weakly stationary.

$$\Rightarrow P_t = t\mu + P_0 + \sum_{i=1}^n \varepsilon_i \quad E(\varepsilon_t) = 0, \quad \text{Var}(\varepsilon_t) = \sigma^2.$$

$$P_t - P_{t-1} = \mu + \varepsilon_t \quad \text{“弱有效”}$$

RW₁ ε_t are iid: ① $E(\varepsilon_t) = \mu$ ② $\text{Var}(\varepsilon_t) = \sigma^2$

③ for any measurable function, $h(\cdot)$ and $g(\cdot)$,

$$\text{cov}(h(\varepsilon_t), g(\varepsilon_{t+j})) = 0 \quad \text{for } j \geq 1 \quad (\text{independent})$$

RW₂: ε_t are independent: ① $E(\varepsilon_t) = \mu_t$ ② $\text{Var}(\varepsilon_t) = \sigma_t^2$

③ For any $\dots$, $\text{cov}(\dots) = 0$ for $j \geq 1$

RW₃: Conditional zero mean $\begin{cases} E(\varepsilon_t | P_{t-1}, P_{t-2}, \dots) = 0 \\ \text{Cov}(\varepsilon_t, g(P_{t-1}, P_{t-2}, \dots)) = 0 \end{cases}$

RW₄: zero covariance: $\text{cov}(\varepsilon_t, \varepsilon_{t-j}) = 0$

No guarantee for: $h(\cdot)$ and $g(\cdot)$, $\text{cov}(h(\varepsilon_t), g(\varepsilon_{t-j})) = 0$

maybe: $\text{cov}(\varepsilon_t^2, \varepsilon_{t-j}^2) \neq 0$

Def:

A martingale is a time-series process P_t that:

$$E[P_{t+1} | P_t, P_{t-1}, \dots] = P_t$$

A martingale difference sequence $\varepsilon_{t+1} = P_{t+1} - P_t$ is:

$$E[\varepsilon_{t+1} | P_t, P_{t-1}, \dots] = 0$$

ε : white Noise

研究 平稳 弱相关时间序列



first to test

• 弱平稳 (宽平稳 / 协方差平稳)

这是实际应用中最常用的定义, 它只要求以下三个条件满足:

1. 均值恒定: 序列的长期平均值是一个常数, 没有上升或下降的趋势。
2. 方差恒定: 序列的波动幅度不随时间变化。
3. 协方差只依赖于时间间隔: 任意两点之间的相关性只取决于它们相距多久, 而不取决于具体在哪个时间点。例如, 今天和昨天的关系, 应该和一个月前某天与其前一天的关系相同。

只有正负有意义, 大小无意义
 $y(j)$ 间隔为 j
 $P(j)$
 $\text{cov}(R_t, R_{t-j})$

弱相关: $\lim_{r \rightarrow \infty} \text{cov}(y_t, y_{t+r}) = 0$
 $E(x_t) = \mu$
 $\text{Var}(x_t) = \gamma_0$

'ACF' 自相关函数

$$\rho_k = \frac{\text{Cov}(x_t, x_{t-k})}{\sqrt{\text{Var}(x_t) \cdot \text{Var}(x_{t-k})}} = \frac{\text{Cov}(y_t, y_{t-k})}{\text{Var}(y_t)} \quad (\text{恒方差})$$

$$\hat{\rho}_k = \frac{\sum_{t=k+1}^n (x_t - \bar{x})(x_{t-k} - \bar{x})}{\sum_{t=1}^n (x_t - \bar{x})^2}$$

"PACF" Partial 偏自相关性

$$\text{Cov}(R_t, R_{t-k} | R_{t-1}, R_{t-2}, \dots, R_{t-k-1})$$

AR

MA

Autoregressive

moving average

(数据具有平稳性、弱相关性)

AR 模型

$$\begin{aligned} \text{AR}(1): \quad y_t &= \phi y_{t-1} + \varepsilon_t \quad (|\phi| < 1) & \varepsilon_t: \text{iid}, E(\varepsilon_t) = 0, \text{Var}(\varepsilon_t) = \sigma^2 \\ &= \dots \\ &= \sum_{j=0}^{\infty} \phi^j \cdot \varepsilon_{t-j} \end{aligned}$$

$$E(y_t) = 0; \quad \text{Var}(y_t) = \sum_{j=0}^{\infty} \phi^{2j} \cdot \sigma^2 = \frac{1}{1-\phi^2} \cdot \sigma^2$$

$$\text{Cov}(y_t, y_{t-k}) = \text{Cov}(\dots y_{t-k} + \sum \varepsilon_{t-j} \phi^j, y_{t-k})$$

$$\begin{aligned} y_t = \phi y_{t-1} + \varepsilon_t &= \phi (\phi y_{t-2} + \varepsilon_{t-1}) + \varepsilon_t = \phi^2 y_{t-2} + \phi \varepsilon_{t-1} + \varepsilon_t \\ &= \phi^k y_{t-k} + \sum_{j=0}^k \varepsilon_{t-j} \cdot \phi^j \end{aligned}$$

$$\downarrow$$
$$= \phi^k \cdot \text{Var}(y_{t-k}) + 0$$

$$\Rightarrow \text{Cov}(y_t, y_{t-k}) = \text{Cov}(y_{t-k}, \phi^k y_{t-k}) = \phi^k \cdot \frac{1}{1-\phi^2} \cdot \sigma^2$$

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \varepsilon_t$$

$$\text{Var}(y_t) = \text{Var}(\phi_1 y_{t-1} + \phi_2 y_{t-2}) + \sigma^2$$

$$= \text{Var}(\phi_1 y_{t-1}) + \text{Var}(\phi_2 y_{t-2}) - 2\text{Cov}(\phi_1 y_{t-1}, \phi_2 y_{t-2})$$

格林函数 (Green Function)

用 $G_i (i=1, 2, 3 \dots)$ 来代替未知的系数, 并用待定系数法简化相应运算.

$$y_t = G_0 \varepsilon_t + G_1 \varepsilon_{t-1} + \dots$$

滞后算子 L (lag operator)

$$L \cdot y_t = y_{t-1} ; \quad L^p y_t = y_{t-p}$$

定义滞后多项式:

$$\phi(L) = 1 - \phi_1 L - \dots - \phi_p L^p$$

P 阶自回归模型写作:

$$\phi(L) \cdot y_t = \varepsilon_t$$

弱平稳与弱相关:

弱平稳 { 计算 $E(y_t)$, 是否随时间变化 (有没有 t)
计算 $Var(y_t)$, $\dots$

弱相关: $cov(y_t, y_{t+j})$, 是否随时间变化, 及是否 $\lim_{j \rightarrow \infty} = 0$.

AR(P)

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + \varepsilon_t$$

于是 AR(P):

$$\phi(L) \cdot y_t = \varepsilon_t$$

对期望: 由于迭代后一定是若干 $\phi^j \varepsilon$ 相加, 于是期望一定为 0.

对方差:
$$\begin{cases} y_t = G(L) \varepsilon_t \\ G(L) = G_0 + G_1 L + G_2 L^2 + \dots \end{cases} \quad (\text{格林函数})$$

联立有:

$$\phi(L) G(L) \varepsilon_t = \varepsilon_t$$

$$\Rightarrow (1 - \sum_{k=1}^p \phi_k L^k) \cdot (\sum_{j=0}^{\infty} G_j L^j) \cdot \varepsilon_t = \varepsilon_t$$

$$\downarrow$$
$$(G_0 + \sum_{j=1}^{\infty} G_j L^j)$$

$$(G_0 - G_0 \sum_{k=1}^p \phi_k L^k + \sum_{j=1}^{\infty} G_j L^j - \sum_{k=1}^p \phi_k L^k \cdot \sum_{j=1}^{\infty} G_j L^j) \varepsilon_t = \varepsilon_t$$

$$(G_0 - G_0 \cdot \sum_{k=1}^P \phi_k L^k + \sum_{j=1}^{\infty} G_j L^j - \sum_{j=1}^{\infty} (\sum_{k=1}^P \phi_k G_j L^{k+j}) \varepsilon_t = \varepsilon_t$$

$$\left\{ G_0 \cdot \phi(L) + \left[\sum_{j=1}^{\infty} (G_j - \sum_{k=1}^P \phi_k G_j L^k) \cdot L^j \right] \right\} \varepsilon_t = \varepsilon_t$$

$$\left\{ G_0 - G_0 \sum_{k=1}^P \phi_k L^k + \left[\sum_{j=1}^{\infty} (G_j - \sum_{k=2}^j \phi'_k G_{j-k}) L^j \right] \right\} \varepsilon_t = \varepsilon_t$$

$$\Rightarrow \left\{ G_0 + \left[\sum_{j=1}^{\infty} (G_j - \sum_{k=1}^j \phi'_k G_{j-k}) L^j \right] \right\} \varepsilon_t = \varepsilon_t$$

$$\text{其中 } \phi'_k = \begin{cases} \phi_k, & k \leq P \\ 0, & k > P \end{cases}$$

$$\text{于是 } \begin{cases} G_0 = 1 \\ G_j = \sum_{k=1}^j \phi'_k G_{j-k}, \quad j \geq 1 \end{cases}$$

$$\text{Var}(y_t) = \sum_{j=0}^{\infty} G_j^2 \sigma^2$$

ACF与PACF计算:

$$\text{Cov}(y_t, y_{t-s}) = E(y_t \cdot y_{t-s})$$

Yule-Walker方程

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + \varepsilon_t$$

两侧同乘 y_{t-h} ($h=0, 1, 2, \dots, p$), 于是

$$y_t \cdot y_{t-h} = \phi_1 y_{t-1} \cdot y_{t-h} + \phi_2 y_{t-2} \cdot y_{t-h} + \dots + \phi_p y_{t-p} \cdot y_{t-h} + \varepsilon_t y_{t-h}$$

$$\text{由于 } E(\varepsilon_t \cdot y_{t-h}) = \begin{cases} \sigma^2 & h=0 \\ 0, & h \neq 0 \end{cases}$$

$$\text{于是 } \begin{cases} \gamma_0 = \phi_1 \gamma_1 + \phi_2 \gamma_2 + \dots + \phi_p \gamma_p + \sigma^2 \\ \gamma_h = \phi_1 \gamma_{h-1} + \phi_2 \gamma_{h-2} + \dots + \phi_p \gamma_{h-p}, \quad h=1, 2, \dots, p \end{cases}$$

写成矩阵形式:

$$\begin{bmatrix} 1 & -\phi_1 & -\phi_2 & \dots & -\phi_p \\ -\phi_1 & 1-\phi_2 & -\phi_3 & \dots & 0 \\ -\phi_2 & -\phi_1-\phi_3 & 1-\phi_4 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -\phi_p & -\phi_{p-1} & -\phi_{p-2} & \dots & 1 \end{bmatrix} \begin{bmatrix} \gamma_0 \\ \gamma_1 \\ \gamma_2 \\ \vdots \\ \gamma_p \end{bmatrix} = \begin{bmatrix} \sigma^2 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

对 $AR(2)$,

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \varepsilon_t$$

$$y_0 = \sigma^2 + \phi_1 y_1 + \phi_2 y_2$$

$$y_t y_{t-1} = \phi_1 y_{t-1} y_{t-1} + \phi_2 y_{t-1} y_{t-2} + \varepsilon_t$$

$$y_1 = \phi_1 y_0 + \phi_2 y_1$$

$$y_2 = \phi_1 y_1 + \phi_2 y_0$$

$$\begin{bmatrix} 1 & -\phi_1 & -\phi_2 \\ -\phi_1 & 1-\phi_2 & 0 \\ -\phi_2 & -\phi_1 & 1 \end{bmatrix} \begin{bmatrix} y_0 \\ y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} \sigma^2 \\ 0 \\ 0 \end{bmatrix}$$

$$\phi_1 y_0 = (1-\phi_2) y_1, \quad y_0 = \frac{1-\phi_2}{\phi_1} y_1$$

$$-\phi_2 y_0 - \phi_1 y_1 + y_2 = \frac{(\phi_2-1)\phi_2}{\phi_1} y_1 - \phi_1 y_1 + y_2 = \frac{(\phi_2-1)\phi_2 - \phi_1^2}{\phi_1} y_1 = -y_2.$$

$$\frac{1-\phi_2}{\phi_1} y_1 - \phi_1 y_1 + \phi_2 \frac{(\phi_2-1)\phi_2 - \phi_1^2}{\phi_1} y_1 = \sigma^2.$$

$$\frac{1-\phi_2 - \phi_1^2 + \phi_2^2(\phi_2-1) - \phi_1^2\phi_2}{\phi_1} = (1-\phi_2)^2(1+\phi_2) - \phi_1^2(1+\phi_2).$$

$$\begin{cases} y_1 = \frac{\phi_1 \sigma^2}{(1+\phi_2)[(1-\phi_2)^2 - \phi_1^2]} \\ y_0 = \frac{(1-\phi_2) \cdot \sigma^2}{\dots} \\ y_2 = [\phi_1^2 + (1-\phi_2)\phi_2] \sigma^2 \\ \dots \end{cases}$$

于是:
$$P_k = \begin{cases} 1, & k=0 \\ \frac{\phi_1}{1-\phi_2}, & k=1 \\ \phi_1 P_{k-1} + \phi_2 P_{k-2}, & k \geq 2. \end{cases}$$

若 $|\phi| < 1$, 就证明了: P_k 与 t 无关且 $\rightarrow 0$.

Equation 3.8.

$$\begin{bmatrix} y_0 & y_1 & \cdots & y_{p-1} \\ y_1 & y_0 & \cdots & y_{p-2} \\ \vdots & \vdots & & \vdots \\ y_{p-1} & y_{p-2} & \cdots & y_0 \end{bmatrix} \begin{bmatrix} \phi_1 \\ \phi_2 \\ \vdots \\ \phi_p \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_p \end{bmatrix}$$

解释: 通过该方程, 若 $\hat{y}$ 得到估计, 则参数 ϕ 可以被确定.

PACF:

$$\rho_{y_t, y_{t-k} | y_{t-1}, \dots, y_{t-k+1}} = \frac{E[(y_t - \hat{E} y_t)(y_{t-k} - \hat{E} y_{t-k})]}{E[(y_{t-k} - \hat{E} y_{t-k})^2]}$$

$$\text{其中 } \hat{E} y_t = E[y_t | y_{t-1}, \dots, y_{t-k+1}]$$

$$\hat{E} y_{t-k} = E[y_{t-k} | y_{t-1}, \dots, y_{t-k+1}]$$

解释: 将已知的部分在 y_t, y_{t-k} 的信息面上的“投影”去除

于是:

$$\hat{E} y_t = \phi_1 y_{t-1} + \cdots + \phi_{k-1} y_{t-k+1} + \phi_k \hat{E} y_{t-k}$$

$$y_t - \hat{E} y_t = \varepsilon_t + \phi_k (y_{t-k} - \hat{E} y_{t-k})$$

$$E[(y_t - \hat{E} y_t)(y_{t-k} - \hat{E} y_{t-k})] = E[\varepsilon_t (y_{t-k} - \hat{E} y_{t-k})] + \phi_k E[(y_{t-k} - \hat{E} y_{t-k})^2]$$

$$= 0 + \phi_k E[\dots]$$

$$\phi_k = \frac{E[(y_t - \hat{E} y_t)(y_{t-k} - \hat{E} y_{t-k})]}{E[(y_{t-k} - \hat{E} y_{t-k})^2]}, \text{ 即 } \rho_{y_t, y_{t-k} | \dots} = \phi_k$$

$$\text{若计算 } \rho_{y_t, y_{t-k-1} | y_{t-1}, \dots, y_{t-k}}, \text{ 则 } y_t - \hat{E} y_t = \varepsilon_t$$

$$\text{此时 } \rho = 0$$

于是: AR(k)模型的PACF只在前k阶不为0, k阶以后全为0, 此即“k阶截尾”

MA (moving average) 模型

$$MA(q): y_t = \varepsilon_t - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2} - \dots - \theta_q \varepsilon_{t-q}$$

定义延后算子 $\theta(L)$: 使 $y_t = \theta(L) \cdot \varepsilon_t$

$$\theta(L) = 1 - \theta_1 L - \dots - \theta_q L^q$$

1. 平稳性与弱相关性

$$E \cdot y_t = 0, \quad D(y_t) = (1 + \theta^2) \cdot \sigma^2$$

$$y_t = \varepsilon_t - \theta_1 \varepsilon_{t-1} - \dots - \theta_q \varepsilon_{t-q}$$

$$\text{Cov}(y_t, y_{t-k}) = \begin{cases} \varepsilon_t, & -\theta_1 \varepsilon_{t-1}, \dots, -\theta_k \varepsilon_{t-k}, \dots, -\theta_q \varepsilon_{t-q} \\ & \varepsilon_{t-k}, \dots, -\theta_{q-k} \varepsilon_{t-q}, \dots, -\theta_q \varepsilon_{t-k-q} \end{cases}$$

$$\text{Cov}(y_t, y_{t-k}) = \left(\sum_{j=1}^{q-k} \theta_j \cdot \theta_{j+k} - \theta_k \right) \cdot \sigma^2$$

当 $k > q$ 时, $\text{Cov}(y_t, y_{t-k}) = 0$;

2. 可逆性

一个 $MA(q)$ 具有可逆性, 若它可以被逆转为一个 $AR(\infty)$

以 $MA(1)$ 为例:

$$y_t = \varepsilon_t - \theta \varepsilon_{t-1}; \quad \varepsilon_t = y_t + \theta \varepsilon_{t-1}$$

$$y_t = \varepsilon_t - \theta (y_{t-1} + \theta \varepsilon_{t-2}) = \varepsilon_t - \theta y_{t-1} - \theta^2 \varepsilon_{t-2}$$

$$= \dots = \varepsilon_t - \theta y_{t-1} - \theta^2 y_{t-2} - \dots - \theta^{t-1} y_1 - \theta^t \varepsilon_0$$

当且仅当 $\theta^t \varepsilon_0 \rightarrow 0$ 时,

$$MA(1) \leftrightarrow AR(\infty): \quad \varepsilon_t = y_t + \theta y_{t-1} + \dots + \theta^{t-1} y_1$$

此时: $|\theta| < 1$.

3. ACF与PACF性质

ACF: 由上述推导, ACF是 q 阶截尾的

PACF: 由于 $AR(p)$ 的PACF p 阶截尾, 而 $MA \rightarrow AR(\infty)$,
于是 MA 的 PACF 不截尾.

ARMA 模型 (自回归移动平均)

$ARMA(p, q)$

平稳性 无偏性

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2}$$

$$y_t - y_{t-1} = (\phi_1 - 1) y_{t-1} + \phi_2 y_{t-2}$$

$$\Delta y_t = (\phi_1 - 1) y_{t-1} + \phi_2 y_{t-2}$$

$$\Delta y_t = (\phi_1 - 1) y_{t-1} + \phi_2 y_{t-1} - \phi_2 y_{t-1} + \phi_2 y_{t-2}$$

$$\Delta y_t = (\phi_1 + \phi_2 - 1) y_{t-1} - \phi_2 \Delta y_{t-1}$$

假设检验

1. likelihood ratio (LR).

用于判断对参数施加的约束是否与数据一致

Is the fit of the unrestricted model significantly better than the restricted?

l_T : log-likelihood function.

$\hat{\theta}$: unrestricted estimator

$\tilde{\theta}$: restricted ~

LR Statistic: $LR = 2[l_T(\hat{\theta}) - l_T(\tilde{\theta})]$

under H_0 , $LR \xrightarrow{d} \chi^2_s$, s : freedom, num of restrictions

2. Lagrange multiplier (LM) also as 'Rao's Score Test'

$$LM = \left[\frac{\partial \log L_T(\theta)}{\partial \theta'} \right]_{\theta=\tilde{\theta}} \cdot \left[- \frac{\partial^2 \log L_T(\theta)}{\partial \theta \partial \theta'} \right]_{\theta=\tilde{\theta}}^{-1} \cdot \left[\frac{\partial \log L_T(\theta)}{\partial \theta} \right]_{\theta=\tilde{\theta}}$$

advantage: only need to estimate the restricted model ($\tilde{\theta}$)

under H_0 , $LM \xrightarrow{d} \chi^2_s$

the score vector: $s(\tilde{\theta}) = \frac{\partial l_T(\theta)}{\partial \theta}$

↓

evaluate how much the likelihood of data changes.

3. Wald Test (W)

If we run our unrestricted model and get our best estimator $\hat{\theta}$;

does this estimate satisfy the restriction $h(\theta) = 0$?

$$W = h'(\hat{\theta}) \cdot [\widehat{\text{Var}}[h(\hat{\theta})]]^{-1} \cdot h(\hat{\theta})$$

$$\widehat{\text{Var}}[h(\hat{\theta})] = \left(\frac{\partial h(\theta)}{\partial \theta'} \right)_{\theta=\hat{\theta}} \left[- \frac{\partial^2 \log L_T(\theta)}{\partial \theta \partial \theta'} \right]_{\theta=\hat{\theta}}^{-1} \left(\frac{\partial h(\theta)}{\partial \theta} \right)_{\theta=\hat{\theta}}$$

$$W \sim \chi^2_s$$

GMM (Generalized Method of Moment)

When the conditions are more than the number of unknown parameters,

$$\min M^T(X, \bar{\theta}) W_{rxr} M(X, \bar{\theta})$$

$$\text{where } M(X, \theta) = \begin{cases} \frac{1}{n} \sum m_1(X, \bar{\theta}) \\ \vdots \\ \frac{1}{n} \sum m_r(X, \bar{\theta}) \end{cases}$$

W_{rxr} : A positive - defined weighting matrix.

The matrix assigned relative importance to each equation.

Estimate the ARMA.

unknown quantities: $\mu, \sigma^2, \phi_1 \sim \phi_p, \theta_1 \sim \theta_q : P+q+2$.

The mean of y_t can be estimated by the sample average; so: $\mu = \bar{y}$.

① AR(p).

Yule-Walker: (Biased in small sample; terrible for MA models)

$$y_t = \phi_1 y_{t-1} + \dots + \phi_p y_{t-p} + \varepsilon_t$$

$$y_t y_{t-1} = \phi_1 y_{t-1}^2 + \dots + \phi_p y_{t-1} y_{t-p} + y_{t-1} \varepsilon_t$$

expectation:

$$y(1) = \phi_1 y(0) + \dots + \phi_p y(p-1)$$

divided by $y(0)$:

$$p(1) = \phi_1 + \dots + \phi_p p(p-1)$$

we can get p equations, then:

$$\begin{bmatrix} p(1) \\ \vdots \\ p(p) \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & p(1) & \dots & p(p-1) \\ p(1) & \ddots & p(1) & \\ & & \ddots & \\ & & & 1 \end{bmatrix}}_{C_p} \begin{bmatrix} \phi_1 \\ \vdots \\ \phi_p \end{bmatrix}$$

(Yule-Walker function)

to get the vector. $(\phi_1, \phi_2, \dots, \phi_p)$,

only to calculate the C_p^{-1}

For MA(1): $y_t = \varepsilon_t - \theta \varepsilon_{t-1}$; $y(0) = E((\varepsilon_t - \theta \varepsilon_{t-1})^2) = E(\varepsilon_t^2 - 2\theta \varepsilon_t \varepsilon_{t-1} + \theta^2 \varepsilon_{t-1}^2) = (1 + \theta^2) \sigma^2$.
squared.

$$\hat{\phi}_{OLS} = \frac{\sum_{t=2}^T y_t y_{t-1}}{\sum_{t=2}^T y_{t-1}^2} \quad \hat{\phi}_{YW}^2 = \frac{\sum_{t=2}^T y_t y_{t-1}}{\sum_{t=1}^T y_t^2} ; \quad \hat{\phi}_{YW} \text{ is always small.}$$

2. estimate by regression: (OLS.)

$$y = X\phi + \varepsilon; \quad y = (y_{1T}, \dots, y_{TT})^T \in \mathcal{R}^{T-P}, \quad X = \begin{pmatrix} y_P & \dots & y_1 \\ \vdots & & \vdots \\ y_{T-1} & & y_{T-P} \end{pmatrix}$$

The OLS estimator is: $\hat{\phi} = (X^T X)^{-1} X^T y$

advantage: ① Lower small-sample bias ② flexibility to add control variables.

But: The formula $(X^T X)^{-1} X^T y$ only works for the model that are **linear in their parameters** and where all variables in the X matrix are directly observed.

For MA(1) model: $y_t = \varepsilon_t - \theta \varepsilon_{t-1}$

$$\therefore \varepsilon_t = y_t + \theta \varepsilon_{t-1} = y_t + \theta (y_{t-1} + \theta \varepsilon_{t-2}) = y_t + \theta y_{t-1} + \theta^2 y_{t-2} + \dots$$

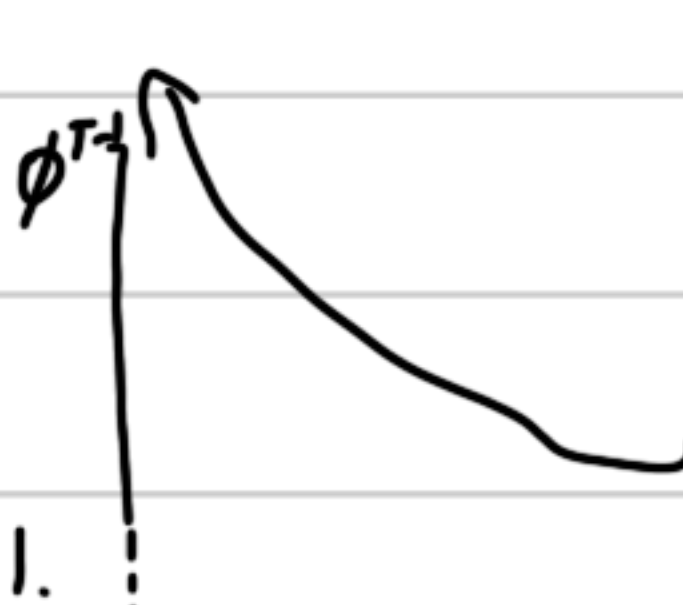
then the θ enters the residual equation non-linearly.

$\sum \varepsilon_t^2$ will have no closed-form matrix solution.

For the AR(1), $\vec{y} = (y_1, \dots, y_T)^T \sim N(0, \Sigma)$

$$\Sigma = \begin{bmatrix} y(0) & y(1) & \dots & y(T-1) \\ y(1) & & & \\ & \ddots & & \\ & & y(0) & \end{bmatrix}, \quad y_t = \phi y_{t-1} + \varepsilon_t, \quad \text{Var.}(y_t) = \phi^2 \text{Var.}(y_{t-1}) + \sigma^2$$

$$\therefore y(0) = \frac{\sigma^2}{1-\phi^2}$$

$$\Sigma = y(0) \cdot \begin{bmatrix} 1 & \phi & & \\ \phi & 1 & & \\ & & \ddots & \\ & & & 1 \end{bmatrix}$$


$$y_t y_{t-1} = \phi y_{t-1}^2 + \varepsilon_t y_{t-1}$$

$$y(1) = \phi \cdot y(0)$$

$$y(2) = \phi y(1) = \phi^2 y(0)$$

$$= \frac{\sigma^2}{1-\phi^2} C$$

3.

MLE.

$$y \sim N(0, \Sigma);$$

By Wold's Decomposition Theorem, $y_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}$ (For any ARMA(p, q))

$$\vec{y} = \begin{pmatrix} 1 & \psi_1 & \psi_2 & \dots \\ 0 & 1 & \psi_1 & \dots \\ 0 & 0 & 1 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} \varepsilon_1 \\ \varepsilon_0 \\ \varepsilon_{-1} \\ \vdots \end{pmatrix} + \begin{pmatrix} 0 & 0 & \dots \\ 1 & 0 & \dots \\ \psi_1 & 1 & \dots \\ \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} \varepsilon_2 \\ \varepsilon_3 \\ \vdots \end{pmatrix}$$

$$\vec{y} = \Psi \cdot \varepsilon^*$$

$$\Sigma = \text{Var.}(y) = \text{Var.}(\Psi \varepsilon^*) = \Psi \text{Var.}(\varepsilon^*) \Psi^T = \sigma^2 \cdot \Psi \cdot \Psi^T$$

$$f(y|\Sigma) = \frac{1}{(2\pi)^{T/2} |\Sigma|^{1/2}} e^{-\frac{1}{2} y^T \Sigma^{-1} y}$$

$\Sigma = \Sigma(\varphi)$; since φ is a vector of unknown parameters.

$$\ell(\varphi) = -\frac{T}{2} \ln 2\pi - \frac{1}{2} |\Sigma(\varphi)| - \frac{1}{2} y^T \Sigma^{-1}(\varphi) y$$

For this function, we want to find the $\max \arg(\varphi)$; that's not easy.

So we can only try to find the numerical solutions.

use conditioning to simplify:

$$f(y_1, y_2, \dots, y_T) = f(y_1) f(y_2|y_1) \dots f(y_T|y_{T-1}, y_{T-2}, \dots, y_1)$$

$$\therefore \ell(y_1, y_2, \dots, y_T, \varphi) = \sum_{t=p+1}^T \ell(y_t|y_{t-1}, y_{t-2}, \dots, y_1) + \ell(y_1, y_2, \dots, y_p)$$

$$y_t|y_{t-1}, \dots, y_1 \sim N(\phi_1 y_{t-1} + \dots + \phi_p y_{t-p}, \sigma^2)$$

only to invert the joint part;

so we only need to invert a $p \times p$ matrix instead of $T \times T$.

General ARMA(p, q)

if MA(q) is invertable;

$$\text{MA(1) for ins: } y_t = \varepsilon_t - \theta \varepsilon_{t-1} = (1 - \theta L) \cdot \varepsilon_t$$

$$\varepsilon_t = \frac{1}{1 - \theta L} y_t$$

$$= [1 + \theta L + (\theta L)^2 + \dots] \cdot y_t$$

$$|\theta| < 1 \rightarrow \text{MA}(q) \text{ is invertable} \rightarrow \text{MA}(q): y_t = c_1 y_{t-1} + c_2 y_{t-2} + \dots + \varepsilon_t$$

then ARMA(p, q) can write as linear combination of y_i

① Truncation: we just has finite data; for data we don't have, set as 0.

② Recursive: ARMA(1,1) for instance;

$$y_t = \phi y_{t-1} + \varepsilon_t - \theta \varepsilon_{t-1}$$

$$\text{set } y_0 = \varepsilon_0 = 0$$

give a set of guesses of parameter: such as $\phi = 0.5$, $\theta = 0.2$.

recurring calculate the residuals:

$$\begin{cases} t=1, \varepsilon_1 = y_1 - \phi y_0 + \theta \varepsilon_0 = y_1 \\ t=2 \\ \vdots \end{cases}$$

then the unobservable residual sequence $(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_T)$ can be calculated by the known y

② Construct the objective function (SSR)

$$SSR(\phi, \theta) = \sum_{t=1}^T \varepsilon_t^2$$

Forecasting:

$$\hat{y}_{T+j|T} = E(y_{T+j} | y_1, y_2, \dots, y_T)$$

Theorem:

$$\hat{y}_{T+j|T} = \arg \min_{h \in \mathcal{H}} E[(y_{T+j} - h(y_1, y_2, \dots, y_T))^2]$$

conditional expectations are the only best predictor with min mean squared error.

In practical forecast, we must use $\hat{\phi}$ replaces ϕ .

$$y_{T+r} - \hat{y}_{T+r|T} = [y_{T+r} - \hat{y}_{T+r}(\phi)] + [\hat{y}_{T+r|T}(\phi) - \hat{y}_{T+r|T}(\hat{\phi})]$$

$$= (\underbrace{\phi^{r-1} \varepsilon_{T+1} + \dots + \varepsilon_{T+r}}_{\text{intrinsic}}) + (\underbrace{\phi^r - \hat{\phi}^r}_{\text{Parameter Estimation Error}}) \cdot y_T$$

forecast interval:

$$\hat{y}_{T+r|T} \pm z_{\alpha/2} \cdot \sqrt{\hat{\sigma}^2 \cdot (1 + \hat{\phi}^2 + \dots + \hat{\phi}^{2r-2})} \quad (\text{only if } \varepsilon \sim N(0, \sigma^2))$$

the interval will not shrink to 0

while $\hat{\phi} \rightarrow \phi$ when n (scale of data) $\rightarrow \infty$

VAR.

definition: a generalization of AR(p) model by adding more time sequences.

① The mean of y_t can be estimated by the sample average, so we consider the demand process only (去中心化)

② **Granger Causality**: 在预测 Y 时, 除了 Y 自身的信息, 加入 X 的过去信息^能 让预测更准
 $\Rightarrow$ 称 X 是 Y 的格兰杰原因 (X Granger-causes Y)

Var (1): $y_t = \phi y_{t-1} + \varepsilon_t, \quad t=1, 2, \dots$

$$\phi = \begin{bmatrix} \phi_{11} & \phi_{12} & \dots & \phi_{1k} \\ \vdots & \vdots & & \vdots \\ \phi_{k1} & \phi_{k2} & \dots & \phi_{kk} \end{bmatrix} \quad \varepsilon_t: k \times 1 \text{ vector.}$$

Assumption:

① $E(\varepsilon_t) = 0$ for all t .

② $E(\varepsilon_t \varepsilon_t') = \Sigma$; where Σ is an $k \times k$ positive definite matrix, (同期相关性)

③ $E(\varepsilon_t \varepsilon_{t'}) = 0$, for all $t \neq t'$ (无自相关性)

Companion Form.

$$\begin{bmatrix} y_t \\ y_{t-1} \\ \vdots \\ y_{t-p+2} \\ y_{t-p+1} \end{bmatrix} = \begin{bmatrix} \phi_1 & \phi_2 & \dots & \phi_{p-1} & \phi_p \\ I_k & & & & \\ & I_k & & & \\ & & \ddots & & \\ & & & I_k & 0 \end{bmatrix} \begin{bmatrix} y_{t-1} \\ y_{t-2} \\ \vdots \\ y_{t-p+1} \\ y_{t-p} \end{bmatrix} + \begin{bmatrix} \varepsilon_t \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix}$$

From a far origin $t = -M+p$. taking y_{-M+p} as given, we have:

$$\begin{aligned} Y_t &= \phi Y_{t-1} + U_t = \phi^2 Y_{t-2} + \phi U_{t-1} + U_t \\ &= \phi^{t+M-p} Y_{-M+p} + \sum_{j=0}^{t+M-p-1} \phi^j U_{t-j} \end{aligned}$$

Selection matrix:

$$Y_t = (I_k \ 0 \ \dots \ 0)_{k \times kp} \cdot Y_t.$$

Therefore, if Y_t is stationary, then y_t will also be stationary.

if all eigenvalues of ϕ lies in the unit circle ($|\lambda| < 1$) and $n \rightarrow \infty$,
and then Y_t converges in mean squared errors to $\sum_{j=0}^{\infty} \phi^j U_{t-j}$

VaR 分析:

① 构建伴随矩阵 (companion matrix)

② 计算特征值 (eigenvalues)

③ stationary

Estimation:

rewrite:
$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + u_t$$
$$= A' g_t + u_t.$$

$$A' = (\phi_1, \phi_2, \dots, \phi_p) \quad g_t = (Y_{t-1}, Y_{t-2}, \dots, Y_{t-p})$$

特征方程的根与伴随矩阵的特征值在数学上互为倒数 $z = \frac{1}{\lambda}$
↓
滞后算子的特征方程

对 rewrite 方程进行 OLS 估计与假设检验的条件:

- ① stationary 特征方程 $|I_k - \phi_1 \lambda - \dots - \phi_p \lambda^p| = 0$ 的根在单位圆外 ($|\lambda| > 1$)
- ② 误差服从多元正态 $u_t \sim N(0, I)$
(计算精确的标准误、t-estimator、F-estimator)
- ③ 自变量间不能有完全共线性 (No perfect Collinearity)
- ④ 初始值给定 (Initial Values Given)

$$\begin{cases} \hat{A}' = \left(\sum_{t=1}^T Y_t g_t' \right) \left(\sum_{t=1}^T g_t g_t' \right)^{-1} \\ \tilde{\Sigma} = T^{-1} \sum_{t=1}^T (Y_t - \hat{A}' g_t) (Y_t - \hat{A}' g_t)' \end{cases}$$

最大似然函数估计值: $l(\hat{A}, \tilde{\Sigma}) = -\frac{Tk}{2} (1 + \log 2\pi) - \frac{T}{2} \cdot \log |\tilde{\Sigma}|$

Order Selection:

$AIC = l_{\max} - \text{参数个数} = l_{\max} - k^2 p = l_{\max} - ks$, 其中 $s = kp$.

$BIC = l_{\max} - \frac{1}{2} \cdot \text{参数个数} \cdot \log(T) = l_{\max} - \frac{ks}{2} \cdot \log(T)$

Choose the max information criterion.

LR (Likelihood Ratio)

$$LR_{p,p} = T \cdot (\log |\hat{\Sigma}_p| - \log |\tilde{\Sigma}_p|)$$

西姆斯修正 (Sims' Correction):

在小样本下, LR 统计量会失真 (过度拒绝)

$$LR^* = (T - kp) \cdot (\log |\tilde{\Sigma}_R| - \log |\tilde{\Sigma}_P|) \quad (\text{自由度调整后的 LR})$$

Tests

块 (Block) 格兰杰非因果检验

将 VAR 写成分块矩阵形式:

$$\begin{cases} Y_1 = G_1 A_{11} + G_2 A_{12} + U_1 \\ Y_2 = G_1 A_{21} + G_2 A_{22} + U_2 \end{cases}$$

$$H_0: A_{12} = 0$$

自由度公式: $df = k_1 \cdot k_2 \cdot p$ $\rightarrow$ 滞后阶数.
限制方程个数 $\downarrow$ 剔除变量个数

$$LR(A_{12}=0) = T \cdot (\log |\tilde{\Sigma}_R| - \log |\tilde{\Sigma}|) \quad \text{或 Sims 修正: } (T - kp) \cdot (\log |\tilde{\Sigma}_R| - \log |\tilde{\Sigma}|)$$

$\downarrow$ 有约束 $\downarrow$ 无约束

块递归 (Block Recursion)

$H_0: A_{12} = 0$ 成立时,

$$\begin{cases} Y_1 = G_1 A_{11} + U_1 \\ Y_2 = G_1 A_{21} + G_2 A_{22} + U_2 \end{cases}$$

实践中的残差计算:

1. 估计包含 4 个变量的完整 VAR(1) 模型; 得到 $|\tilde{\Sigma}|$

$$\tilde{\Sigma} = \frac{1}{n} \sum_{t=1}^n \hat{u}_t \hat{u}_t'$$

2. 估计受约束模型; 三步投影法:

① 单独用 1 日变量对自身滞后项运行 OLS, 得 $\hat{u}_1$

$$\begin{bmatrix} y_{1,t} \\ y_{2,t} \end{bmatrix} = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix} \begin{bmatrix} y_{1,t-1} \\ y_{2,t-1} \end{bmatrix} + U_t$$

② 用新变量对所有滞后项与 G_1 运行 OLS, 得 $\hat{u}_2$

$\downarrow$
两个估计, 对每个 t , 得 u_1, u_2

③ 合并残差; 计算 $|\tilde{\Sigma}_R|$

$\downarrow$
用 Y_2 的两个变量分别对四个变量与 $\hat{u}_1$ 回归

注意: $\hat{u}$ 是一个 $k \times t$ 的矩阵!

每一期, $y_{1,t}$ $y_{2,t}$ 都会有残差。

Forecast:

如何对未来 h 期的多变量 $y_{T+h|T}$ 进行预测?

$$\text{求 } \hat{y}_{T+h|T} = E(y_{T+h} | \Omega_T), \Omega_T \text{ 为信息集}$$

多维预测中 e 是一个向量;

$$\text{二次损失函数: } L_q = (y_{T+h} - \hat{y}_{T+h|T})' A (y_{T+h} - \hat{y}_{T+h|T})$$

A : 对称正定矩阵; 代表不同变量预测误差的相对重要性

The optimal forecast does not depend on A .

$E(y_{T+h} | \Omega_T)$ 永远是最优估计

$$\text{在 Var(1) 中, } \hat{y}_{T+h|T} = \phi^h y_T$$

Recursive Forecasting:

$$\hat{y}_{T+j|T} = \sum_{i=1}^p \phi_i \cdot \hat{y}_{T+j-i|T}, j=1, 2, \dots, h$$

实例:

Var(2) 预测三期;

$$\hat{y}_{T+1|T} = \phi_1 y_T + \phi_2 y_{T-1}$$

$$\hat{y}_{T+2|T} = \phi_1 \hat{y}_{T+1|T} + \phi_2 y_T$$

即: 不断用预测值代替

预测区间:

$$\text{取决于误差项累积: } \sum_{j=0}^{h-1} \phi^j \cdot u_{t-j}$$

SVAR. (structured.)

Var 缺陷: 右边只有滞后项, 同期变量间无法相互影响

↓

全部被归入 u_t 中

u_t 的 Σ 非对角线不为 0, 即冲击间强相关

无法识别单一政策的经济冲击

$$B y_t = \phi_1 y_{t-1} + \dots + \phi_p y_{t-p} + u_t$$

B : $k \times k$ 代表了同期变量间的相互影响

三角化约束与乔里斯基分解:

① B 是下三角矩阵 (递归的因果链条)

② 结构冲击: 所有同期相关性提取后 u_t 会完全互不相关; Σ' 是对角矩阵

↓ 称为结构冲击

③ 乔里斯基分解: 将非对角协方差矩阵, 分解为三角矩阵 B 与对角协方差矩阵 Σ

1. 建立 SVAR 模型:

$$y_{1t} = \gamma_{10} - b_{12} y_{2t} + \gamma_{11} y_{1,t-1} + \gamma_{12} y_{2,t-1} + \varepsilon_{1t} \quad (1)$$

$$y_{2t} = \gamma_{20} - b_{21} y_{1t} + \gamma_{21} y_{1,t-1} + \gamma_{22} y_{2,t-1} + \varepsilon_{2t} \quad (2)$$

2. 施加识别约束 $b_{12} = 0$

$$y_{1t} = \gamma_{10} + \gamma_{11} y_{1,t-1} + \gamma_{12} y_{2,t-1} + \varepsilon_{1t} \quad (3)$$

3. 求解估计 Reduced Form:

将 (3) 代入 (2), 得到 y_{2t} 只关于滞后项的表达

$$y_{1t} = \alpha_0 + \alpha_1 y_{1,t-1} + \alpha_2 y_{2,t-1} + u_{1t},$$

与 (3) 共同进行 OLS 估计, 得

$$u_{1t}, u_{2t}.$$

$$\text{求 } \hat{\Sigma} = \begin{pmatrix} \hat{\sigma}_{u_1}^2 & \hat{\sigma}_{u_{12}} \\ \hat{\sigma}_{u_{12}} & \hat{\sigma}_{u_2}^2 \end{pmatrix}.$$

4. 反向解出结构参数

脉冲响应函数 (IRF)

$$I_y(n, \delta, \mathcal{I}_{t-1}) = \underbrace{E(y_{t+n} | u_t = \delta, u_{t+1} = 0, \dots)}_{\text{实验组}} - \underbrace{E(y_{t+n} | u_t = 0, u_{t+1} = 0, \dots)}_{\text{对照组}}$$

n : 期数.

δ : 脉冲大小.

$\mathcal{I}_{t-1}$: 历史信息集.

对 $VAR(1)$:

若 $VAR(1)$ stationary, 则

$$\begin{aligned} VAR(1) &= u_t + \phi u_{t-1} + \phi^2 u_{t-2} + \dots \\ &= \sum_{i=0}^{\infty} \phi^i u_{t-i} \end{aligned}$$

$$\therefore I_y(n, \delta, \mathcal{I}_{t-1}) = A_n \delta = \phi^n \delta.$$

$VAR(p)$:

$$VAR(p) = \sum_{i=0}^{\infty} A_i u_{t-i}$$

对 $VAR(p)$ 有 Recursion: $A_j = \phi_1 A_{j-1} + \phi_2 A_{j-2} + \dots + \phi_p A_{j-p}$

$$A_0 = I_k; \quad \text{当 } j < 0 \text{ 时, } A_j = 0$$

正交化脉冲函数:

① 由 cholesky 分解,

对称正定矩阵

reduced OLS 残差协方差矩阵 $\Sigma = PP'$, 其中 P 是三角矩阵

$$\begin{aligned} \textcircled{2} \quad y_t &= \sum_{j=0}^{\infty} A_j u_{t-j} = \sum_{j=0}^{\infty} A_j (PP') u_{t-j} \\ &= \sum_{j=0}^{\infty} (A_j P) (P' u_{t-j}), \end{aligned}$$

③ 定义: $\eta_t = P' u_t$, $B_j = A_j P$

$$\therefore y_t = \sum_{j=0}^{\infty} B_j \eta_{t-j}$$

$\begin{cases} B_j: \text{正交化脉冲响应系数} \\ \eta_j: \text{正交化冲击} \end{cases}$

可以证明: $\text{Var}(\eta_j) = I_k$

正交脉冲响应函数:

$$B_j = A_j P \quad (\text{下三角矩阵 } P \text{ 的顺序尤其重要})$$

A_j : 诱导型脉冲响应矩阵

实战计算:

- {

① 求 reduced 的 Σ

② 解出 P

③ 由 $B = P^{-1}$, 得 SVAR (注意标准化为对角线为1)

④ 由 $\text{Var}(P)$, 计算 Recursion 的 A_i

区分 B 与 B_i : B 是 SVAR 的同期响应矩阵,
 $By_t = \dots + \epsilon_t$;
 $B = P^{-1}$

② B_i 是 IRF 的动态变量

$A_j = \phi_1 A_{j-1} + \phi_2 A_{j-2} + \dots + \phi_p A_{j-p}$
 $A_0 = I_k, A_j = 0, j < 0$
 ϕ_i 为滞后项系数矩阵

选择向量: $e_i = \epsilon_i$, $B_n e_i = B_n$ 的第 i 列

$$\begin{aligned} \text{双夹逼公式: } OI_{ij,n} &= e_j' A_n P e_i \\ &= e_j' B_n e_i \end{aligned}$$

即 B_n 的 (j, i) 的元素

第 i 个变量的第 j 个值; 第 i 个变量受到冲击后导致第 j 个变量受到的影响

Cointegration (协整)

Integrated (单整): $I(d)$: 对一个时间序列进行 d 阶差分后, 才能平稳.

伪回归问题: 由于共同的时间趋势, 导致不相关的两个变量
呈现高度显著的虚假关系;

识破伪回归: 加入时间趋势项 t

协整: 两个或多个本身非平稳 ($I(1)$) 的时间序列, 若存在长期、稳定的均衡关系,
称它们之间存在协整关系.

ARDL (autoregressive Distributed Lag Model)

自回归分布滞后模型

$$y_t^* = \alpha + \beta x_t + u_t, \quad y_t^* \text{ 为理想水平}$$

$$\text{而 } y_t - y_{t-1} = \lambda(y_t^* - y_{t-1}), \quad \lambda \in (0, 1)$$

$$\begin{aligned} y_t &= (1-\lambda) \cdot y_{t-1} + \lambda y_t^* \\ &= \underbrace{(1-\lambda)y_{t-1}}_{\text{自回归部分}} + \underbrace{\lambda\alpha + \lambda\beta x_t + \lambda u_t}_{\text{分布滞后部分}} \end{aligned}$$

优势: 可以同时得出“短期效应”与“长期效应”

$$\left\{ \begin{array}{l} \text{短期效应: } \frac{\partial y_t}{\partial x_t} = \lambda\beta \\ \text{长期效应: 当长期调整后, } y_t = y_{t-1} = \tilde{y}, \quad x_t = \tilde{x} \end{array} \right.$$

$$\tilde{y} = (1-\lambda)\tilde{y} + \lambda\alpha + \lambda\beta\tilde{x} + \lambda\alpha$$

$$\therefore \tilde{y} = \alpha + \beta\tilde{x} + \tilde{u}$$

$$\frac{\partial \tilde{y}}{\partial \tilde{x}} = \beta$$

ARDL(p, q): (p : 自回归阶数; q : 分布滞后阶数)

$$y_t + \lambda_1 y_{t-1} + \dots + \lambda_p y_{t-p} = \alpha + \beta_0 x_t + \beta_1 x_{t-1} + \dots + \beta_q x_{t-q} + u_t$$

用滞后算子 L 写为紧凑形式:

$$\lambda(L)y_t = \alpha + \beta(L)x_t + u_t$$

$$\text{平稳条件: } \lambda(z) = 1 + \lambda_1 z + \lambda_2 z^2 + \dots + \lambda_p z^p = 0$$

特征根 z 必须在单位圆外部

ARDL

VAR

单方程模型

多方程联立系数

内生 y , 外生 x

所有变量内生

单向传导

存在双向反馈

参数少

参数极多

估计方法:

估计一致性的条件: ① 自变量间不能有完全共线性 (可选)

② 自变量与误差项在长期不相关

直接用 OLS 估计即可: $\hat{\theta} = (Z^T Z)^{-1} \cdot Z^T y$

ECM (Error Correction Model, 误差修正模型)

ARDL(1,1)

$$y_t = \alpha + \beta_0 x_t + \beta_1 x_{t-1} + \lambda y_{t-1} + u_t$$

$$\Delta y_t = \alpha + \beta_0 x_t + \beta_1 x_{t-1} + (\lambda - 1) y_{t-1} + u_t$$

$$\Delta y_t = \alpha + \beta_0 \Delta x_t + (\beta_0 + \beta_1) x_{t-1} + (\lambda - 1) y_{t-1} + u_t$$

$$= \alpha + \beta_0 \Delta x_t - (1 - \lambda) \cdot (y_{t-1} - \frac{\beta_0 + \beta_1}{1 - \lambda} x_{t-1}) + u_t$$

$$\text{令 } \theta = \frac{\beta_0 + \beta_1}{1 - \lambda}$$

$$\Delta y_t = \alpha + \beta_0 \Delta x_t - (1 - \lambda) \cdot (y_{t-1} - \theta x_{t-1}) + u_t$$

此即误差修正模型

$$\theta = \frac{\beta_0 + \beta_1}{1 - \lambda} \quad \text{即 } y \text{ 与 } x \text{ 的长期均衡斜率系数}$$

当均衡态 $\Delta y_t = \Delta x_t = 0$ 时,

$$\text{必定有 } \bar{y} = \text{常数} + \theta \bar{x}$$

Q: 什么叫误差修正模型?

令 $e_{t-1} = y_{t-1} - \theta x_{t-1}$, 代表昨天偏离长期均衡的误差

$-(1 - \lambda)$: 误差修正速度

若 y_{t-1} 偏高, $-(1 - \lambda) < 0$, $\therefore \Delta y_t < 0$,

反之, $\Delta y_t > 0$.

Single Cointegration.

Engle-Granger Residual Test. (格格尔-... 两步法协整检验)

1. 进行OLS回归: 在k个非平稳变量里, 挑出 y_{it} 作为因变量
(协整回归)
(Cointegrating Regression) 对剩下的所有变量进行OLS回归
2. 对残差序列 $\hat{\varepsilon}_t$ 进行DF单位根检验

$$H_0: \hat{\varepsilon}_t \text{ 是 } I(1)$$

若拒绝 H_0 , 说明存在协整关系

注意: ① 不能用标准DF临界值. (OLS回归会抹平差异?)

应使用更严格的 Engle-Granger 临界表

② 非对称; 对因变量选择敏感

③ 无法识别多于1个的协整变量

ARDL四界协整检验法 (PSS检验)

1. 估计“无约束误差修正模型”(UECM)

$$\Delta y_t = a_0 + \underbrace{\sum_{i=1}^p \psi_i \Delta y_{t-i} + \sum_{i=0}^p \phi_{i1} \Delta x_{1,t-i} + \sum_{i=0}^p \phi_{i2} \Delta x_{2,t-i}}_{\text{短期波动}} + \underbrace{\delta_1 y_{t-1} + \delta_2 x_{1,t-1} + \delta_3 x_{2,t-1} + u_t}_{\text{长期均衡}}$$

2. 进行联合F检验 (或 Wald 检验)

$$H_0: \delta_1 = \delta_2 = \delta_3 = 0$$

3. 决策

由于同时允许 $I(0)$ 与 $I(1)$ 出现, F统计量不再服从标准F分布

{ 下边界 F_L : 假定所有自变量均为 $I(0)$
上边界 F_U : 所有变量为 $I(1)$

① 若 $F > F_U$: 拒绝 H_0 , 存在协整

② $F < F_L$: 不存在

③: $F_L < F < F_U$: 灰色地带

向量误差修正模型 (VECM)

$$\text{Var}(p): y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + u_t$$

特征: 全部 y_t 非平稳

方法: $y_t = \Delta y_t + y_{t-1}$

$$y_{t-1} = y_{t-1}$$

$$y_{t-2} = y_{t-1} - \Delta y_{t-1}$$

$$y_{t-3} = y_{t-2} - \Delta y_{t-2} = y_{t-1} - \Delta y_{t-1} - \Delta y_{t-2}$$

$\vdots$

$$y_{t-p} = y_{t-1} - \Delta y_{t-1} - \dots - \Delta y_{t-p+1}$$

$$\text{VECM: } \Delta y_t = \underbrace{-\pi y_{t-1}}_{\text{长期修正项}} + \underbrace{\sum_{j=1}^{p-1} \Gamma_j \Delta y_{t-j}}_{\text{短期动态波动项}} + u_t$$

$$\pi = I_k - \phi_1 - \phi_2 - \dots - \phi_p \quad (\text{协整矩阵})$$

Γ_j : 原本参数矩阵的部分求和

$$\Gamma_j = -\sum_{i=j+1}^p \phi_i$$

格兰杰表述定理: $\pi = \alpha \beta'$

设 π 的秩为 r , 则 π 可被唯一分解为两个大小为 $k \times r$ 的矩阵之积

α : 误差修正速度 (载荷矩阵)

β : 长期均衡关系 (协整向量)

如果没有约束关系, π 是 $k \times k$ 矩阵: 无约束参数 $= k^2$

若系统中存在 r 个协整关系, $\text{Rank}(\pi) = r$.

α, β 各有 rk 个参数;

由于可以对它们进行任意伸缩变换; 其中有 r^2 个参数是冗余的

必须施加 r^2 个标准化约束

$$\text{实际参数个数} = kr + kr - r^2 = 2kr - r^2$$

$$\text{实际约束} = k^2 - (2kr - r^2) = (k-r)^2$$

$$L(\theta) = -\frac{1}{2} \sum (\log(\sigma_t^2) + \frac{r_t^2}{\sigma_t^2})$$

LR检验: $l(\theta; r) = -\frac{Tk}{2} \log 2\pi - \frac{T}{2} \log |\Sigma| - \frac{1}{2} \sum_{t=1}^T u_t' \Sigma^{-1} u_t$

其中 $u_t = \Delta y_t + \Pi y_{t-1}$

条件: ① 历史变动值 y_0 为常数.

② u_t 是 iid 的多元正态向量; 协方差向量 Σ 是对称正定的

估计 VECM: 浓缩似然

① 假设 β 已知: $W_{t+1} = \beta' y_{t-1}$

$\therefore \Delta y_t = -\alpha W_{t+1} + u_t$

最小二乘法得

$\hat{\alpha}(\beta) = -S_{01} \beta (\beta' S_{11} \beta)^{-1}$

其中 $\begin{cases} S_{01} = \frac{1}{T} \sum_{t=1}^T \Delta y_t y_{t-1}' \\ S_{11} = \frac{1}{T} \sum_{t=1}^T y_{t-1} y_{t-1}' \end{cases}$

② 代回 u_t 中, 得 $u_t(\hat{\alpha}) = \Delta y_t - S_{01} \beta (\beta' S_{11} \beta)^{-1} \beta' y_{t-1}$

定义 S_{00} : Δy_t 自身的样本协方差矩阵

$\hat{\Sigma}(\beta) = S_{00} - S_{01} \beta (\beta' S_{11} \beta)^{-1} \beta' S_{01}'$

③ 代回 LR 函数中得

$l^c(\beta, r) = -\frac{Tk}{2} (1 + \log 2\pi) - \frac{T}{2} \log |S_{00} - S_{01} \beta (\beta' S_{11} \beta)^{-1} \beta' S_{01}'|$

寻找最完美的长期协整矩阵 β , 等价于求解

$M = S_{10} S_{00}^{-1} S_{01} S_{11}^{-1}$

的特征值.

求出 $k \times k$ 矩阵的 k 个特征值; 从大到小排序:

$\hat{\lambda}_1 > \hat{\lambda}_2 \dots > \hat{\lambda}_k$

λ_i 称为典型相关系数; 特征值越接近 1, 协整关系越明显

若 $M \hat{v}_i = \hat{\lambda}_i \hat{v}_i$,

特征向量 $\hat{v}_1, \dots, \hat{v}_r$, 即长期协整向量 (前 r 个最大特征值)

$\therefore \hat{\beta}_{MLE} = [\hat{v}_1, \hat{v}_2, \dots, \hat{v}_r]$

$\therefore l^c(r) = -\frac{Tk}{2} (1 + \log 2\pi) - \frac{T}{2} \log |S_{00}| - \frac{T}{2} \sum_{i=1}^r \log(1 - \hat{\lambda}_i)$

Johansen 迹检验:
$$Q_{Trace}(r) = -T \sum_{i=r+1}^k \log(1 - \hat{\lambda}_i)$$

序贯检验: 1. 测试 $H_0: r=0$; 若拒绝, 说明 $r \geq 1$,

↓
2. 测试 $H_0: r=1, 2, \dots$ 直到无法拒绝

法1: 最大特征值统计量: H_0 : 系统有 r 个协整 H_1 : 系统有 $r+1$ 个协整

每次只测试下一个特征值 $\hat{\lambda}_{r+1}$ 是否显著

$LR(H_r | H_{r+1}) = -T \log(1 - \hat{\lambda}_{r+1})$; 必须查专门的 Johnson 临界值表.

法2: 迹统计量 $H_0: r$ 个 H_1 : 系统满秩

测试剩下 $k-r$ 个特征值对数累加和

$$LR(H_r | H_k) = -T \sum_{i=r+1}^k \log(1 - \lambda_i)$$

Seasonality

Cramer 分解定理: 任何非平稳的时间序列 x_t , 都可拆为两部分:

$$x_t = \underbrace{\sum_{j=0}^d \beta_j t^j}_{\text{确定性趋势}} + \underbrace{\psi(L) a_t}_{\text{平稳随机波动}}$$

时间序列分解法: T : 趋势 S : 季节 C : 循环 I : 随机

Smoothing the data:

$$M(x_t) = \sum_{i=-k}^f \theta_i x_{t+i}, \quad \forall k, f > 0$$

$$\textcircled{1} \sum_{i=-k}^f \theta_i = 1$$

$$\textcircled{2} k=f, \text{ 且 } \theta_i = \theta_{-i} \text{ (对称性)}$$

时间跨度 n 为奇数, $M_n(x_t) = \sum_{i=-k}^k \frac{x_{t+i}}{n}$

n 为偶, 使用双重移动平均;

$$\begin{aligned} \text{如 } M_{2 \times 4}(x_t) &= \frac{1}{2} M_4(x_t) + \frac{1}{2} M_4(x_{t+1}) \\ &= \frac{1}{2} \left(\frac{x_{t-2} + x_{t-1} + x_t + x_{t+1}}{4} \right) + \frac{1}{2} \left(\frac{x_{t+1} + x_{t+2} + x_{t+3} + x_{t+4}}{4} \right) \end{aligned}$$

对 Moving Average 模型,

$$x_t = a + bt + \varepsilon_t$$

$$E[M(x_t)] = E(x_t)$$

ARIMA.

差分可以消除趋势, 使序列平稳

ARIMA(p, d, q): 其中 d: 差分 d 次可以使序列平稳

$$\Delta^d x_t = (1-L)^d x_t = \sum_{i=0}^d (-1)^i C_d^i x_{t-i}$$

$$\phi(L) \cdot \Delta^d x_t = \theta(L) \cdot \varepsilon_t \quad \left(\begin{array}{l} \Delta^d = (1-L)^d, \phi(L) = 1 - \phi_1 L - \dots - \phi_p L^p \\ \theta(L) = 1 - \theta_1 L - \dots - \theta_q L \end{array} \right)$$

$\phi(L) \Delta^d$: 广义自回归多项式

• ARIMA 的非平稳性来自于 $(1-L)^d$, 即 Δ^d .

ARIMA 的思路是“差分”来消除不平稳性; 但差分本身会损失信息; 高阶差分难以解释

Time Trend

线性: $x_t = a_0 + a_1 \frac{t}{T} + \varepsilon_t$

直接用 OLS 估计

非线性: $x_t = a_0 + a_1 m(\frac{t}{T}) + \varepsilon_t$

其中 $m(t)$ 是一个任意函数

如 $m(t) = (\frac{t}{T} - \mu)^2$ / $m(t) = (t - \mu)^4$

Quadratic 时间趋势

$$x_t = a_0 + a_1 \left(\frac{t}{T} - \mu\right)^2 + \varepsilon_t, \quad \mu \text{ 是转折点}$$

但 OLS 不能估计; 于是写成:

$$x_t = a \cdot \left(\frac{t}{T}\right)^2 + b \left(\frac{t}{T}\right) + c$$

μ 的最优估计为: $-\frac{b}{2a}$

问题: a, b 都是估计量, $\hat{\mu} = -\frac{\hat{b}}{2\hat{a}}$ 并不准确, 会随机波动

Delta Method:

$$\sqrt{n}(\hat{\beta} - \beta) \rightarrow N(0, \Sigma)$$

对一个函数 $g(\cdot)$, 有:

$$\sqrt{n}(g(\hat{\beta}) - g(\beta)) \rightarrow N(0, \Delta g^T \Sigma \Delta g)$$

Δ : 梯度算子.

$$\text{于是有: } \sqrt{n} \cdot \left(-\frac{\hat{b}}{2a} + \frac{b}{2a}\right) \rightarrow N\left(0, \left(-\frac{1}{2a}, \frac{b}{2a^2}\right) \Sigma \begin{pmatrix} -\frac{1}{2a} \\ \frac{b}{2a^2} \end{pmatrix}\right)$$

Grid Search (网格搜索)

对高次曲线, 无法 OLS. $X_t = \beta_0 + \beta_1 \left(\frac{t}{T} - \mu\right)^4 + \varepsilon_t$

① 选择一个认为的对称轴所在区间 (k_0, k_1)

② $\mu = k_0 + \delta i$, 在区间中不断取 μ 的值

③ μ 的值取定时, 可以 OLS.

$$\text{计算 } MSE = \frac{1}{T} \sum \hat{\varepsilon}_t^2;$$

选择 MSE 最小的 μ

④ 固定 μ 后, OLS 检验 β_1 是否显著

Seasonality (季节性)

长期趋势 $\rightarrow$ 周期重复.

加法模型: 季节波动大小固定.
乘法 ~: 季节波动随水平变化

加法模型:

① 先去趋势, $y_t = x_t - T_t = S_t + I_t$

② $y_{i,j} = \bar{y} + S_j + I_{ij}$
期数 $\leftarrow \downarrow$ 均值
 周期的第 j 部分

$$\bar{y} = \frac{\sum \sum y_{ij}}{k \cdot J}, \quad k: \text{周期数}; J: \text{每周期季节数}$$

$$\textcircled{3} \bar{y}_j = \frac{\sum_{i=1}^k y_{ij}}{k}$$

④ $S_j = \bar{y}_j - \bar{y}$: 季节效应的估计

乘法模型:

① 去趋势. $y_t = x_t / T_t$

$$\textcircled{2} \text{求 } \bar{y} = \frac{\sum \sum y_{ij}}{k \cdot J}$$

③ $\bar{y}_j$ 同上

④ $S_j = \bar{y}_j / \bar{y}$

为什么去季节性?

季节性会掩盖真实趋势

去掉季节性后更容易看到真实趋势和非季节变化

$$x_t = \underbrace{a_0}_{\text{基础水平}} + \underbrace{a_1 \left(\frac{t}{T} - \mu\right)^2}_{\text{时间趋势}} + \underbrace{\sum_{j=1}^6 \beta_j D_j}_{\text{季节性}} + \varepsilon_t$$

D_j : dummy variable; $D_1 = \begin{cases} 1, \text{同} \\ 0, \text{other} \end{cases}$

Q: 为什么一周七天, 只有6个D?

(Dummy Variable Trap) $D_1 + D_2 + \dots + D_7 = 1$, 那么

α 与 $D_1 + \dots + D_7$ 完全共线, OLS崩了

于是我们保留 α , 去掉一个 Dummy; 失去的自动成为“基准组”

此时, β_i 表示的是 i 相对于基准组的平均差异

Event Study:

① 定义事件

② 样本选择

③ 正常收益与异常收益 $AR_t = R_t - E(R_t)$ [异常收益]

④ 估计正常收益

⑤ 统计检验

⑥ 实证结果

⑦ 解释与结论

event 类型 $\left\{ \begin{array}{l} a. \text{immediate and permanent 立即永久的} \\ b. \text{underreaction 反应不足 (漂移)} \\ c. \text{overreaction 过度反应} \end{array} \right.$

market model.

$$R_t = \alpha + \beta R_{mt} + \varepsilon_t$$

超额收益能力

市场影响

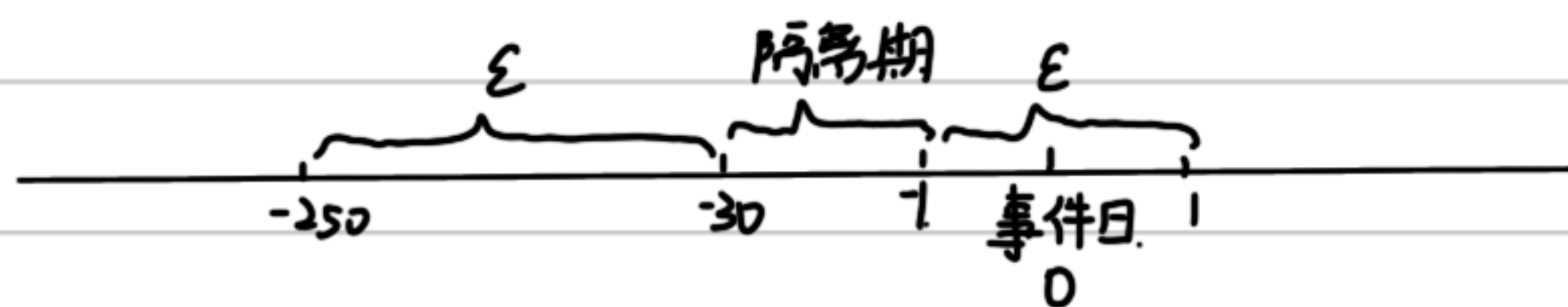
个股特有冲击

β : 衡量系统性风险, 股价对市场冲击的敏感度

事件窗口 ε . 长度 ε 取决于研究目的 长期/短期

估计窗口 W . 用于估计 $\hat{\alpha}, \hat{\beta}$; 尽量长, 但太长会过拟合

Quarantine 隔离期 事件的信息可能提前出现; 加入估计窗口会污染 W .



用估计窗口算 $\hat{\alpha}, \hat{\beta} \rightarrow$ 计算事件发生后的正常收益 $R_t \rightarrow$ 比较 R_t 与实际收益: $AR_t = R_t - (\hat{\alpha} + \hat{\beta}R_{mt})$

$$CAR \text{ (累计异常收益)} = \sum_{t=1}^T AR_t$$

判断 AR 是否显著:

H_0 : 事件没有影响. $AR_t = \varepsilon_t \sim N(0, \sigma^2)$

$$CAR = \sum_{t=1}^T AR_t \sim N(0, T\sigma^2) \quad (\text{或 } CAR(T) = \sum_{t=1}^T \hat{\varepsilon}_t)$$

$$\text{令 } Z = \frac{CAR(T)}{\sigma\sqrt{T}}, \text{ 则 } Z \sim N(0, 1)$$

(SCAR)

$$\text{拒绝域: } |Z| > 1.96 (z_{0.05})$$

$$\text{即 } |CAR(T)| > 1.96\sigma\sqrt{T}$$

$$\text{实际估计中, } SCAR = \frac{CAR(T)}{\hat{\sigma}\sqrt{T}} \sim t(T-2)$$

$\hat{\alpha}, \hat{\beta}$ 的估计方法: OLS.

$$R = X\theta + \varepsilon, \text{ 其中 } R = \begin{bmatrix} r_1 \\ \vdots \\ r_T \end{bmatrix}, X = \begin{bmatrix} 1 & r_{m1} \\ \vdots & \vdots \\ 1 & r_{mT} \end{bmatrix}, \theta = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

$$\therefore \hat{\theta} = (X^T X)^{-1} X^T R$$

$$\hat{\sigma}^2 = \frac{\hat{\varepsilon}^T \hat{\varepsilon}}{T-2}, \quad \hat{\varepsilon} = R - X\hat{\theta}$$

记 R^*, X^* 为事件窗口的数据

$$\hat{\theta} = (X^T X)^{-1} X^T X \theta + (X^T X)^{-1} X^T \varepsilon$$

$$= \theta + (X^T X)^{-1} X^T \varepsilon$$

$$\therefore \hat{\theta} - \theta = (X^T X)^{-1} X^T \varepsilon$$

$$\hat{\varepsilon}^* = R^* - X^* \hat{\theta} = R^* - X^* \theta - X^* (\hat{\theta} - \theta)$$

$$= \varepsilon^* - X^* (X^T X)^{-1} X^T \varepsilon$$

$$\therefore E(\hat{\varepsilon}^* | X^*, X) = 0$$

$$Var(\hat{\varepsilon}^*) = \sigma^2 (I_E + X^* (X^T X)^{-1} X^{*T})$$

$$\text{当 } T \rightarrow \infty \quad \hat{\varepsilon}^* \sim N(0, \sigma^2)$$

Volatility & garch (波动率)

衡量风险

ARCH

$$\text{ARCH}(1): r_t = \sigma_t \varepsilon_t$$

$$\sigma_t^2 = w + d r_{t-1}^2$$

$$\textcircled{1} \varepsilon_t \sim \text{IID } N(0, 1)$$

$$\textcircled{2} E(r_t) = E(E(r_t | F_{t-1})) = E(\sigma_t E(\varepsilon_t)) = 0$$

$$\textcircled{3} E(r_t^2) = E(\sigma_t^2) = w + d E(r_{t-1}^2)$$

$$E(r_{t-1}^2) = E(\sigma_{t-1}^2). \text{ 由平稳性, } E(\sigma_t^2) = E(\sigma_{t-1}^2) = v$$

$$\therefore v = w + d v, \quad v = \frac{w}{1-d}$$

$$\therefore \text{Var}(r_t) = E(r_t^2) - E^2(r_t) = E(r_t^2) = E(\sigma_t^2) = \frac{w}{1-d}$$

$$\text{Var}(r_t) = \frac{w}{1-d}$$

平稳条件: $0 \leq d < 1$

GARCH(1,1)

$$\sigma_t^2 = w + \beta \sigma_{t-1}^2 + \gamma r_{t-1}^2$$

$\swarrow$ $\downarrow$ $\downarrow$
 长期均值 记忆效应 冲击效应

$$r_t = \sigma_t \varepsilon_t$$

$$r_t^2 = \sigma_t^2 \varepsilon_t^2$$

$$\therefore E(r_t^2) = E(\sigma_t^2)$$

$$E(\sigma_t^2) = w + \beta E(\sigma_{t-1}^2) + \gamma E(r_{t-1}^2)$$

$$\therefore E(\sigma_t^2) = \frac{w}{1-\beta-\gamma}$$

平稳条件: $0 < \beta + \gamma < 1$

Garch 本身是“厚尾分布”而非标准的正态

$$\sigma_t^2 = w + \beta \sigma_{t-1}^2 + \gamma r_{t-1}^2$$

$$= \frac{w}{1-\beta} + \gamma \cdot \sum_{j=1}^{\infty} \beta^{j-1} r_{t-j}^2$$

$$r_t^2 = \sigma_t^2 \varepsilon_t^2$$

$$= \sigma_t^2 + \sigma_t^2 (\varepsilon_t^2 - 1)$$

$$= w + \beta \sigma_{t-1}^2 + \gamma r_{t-1}^2 + \sigma_t^2 (\varepsilon_t^2 - 1)$$

$$= w + (\beta + \gamma) r_{t-1}^2 + \beta (6_{t-1}^2 - r_{t-1}^2) + 6_{t-1}^2 (\varepsilon_t^2 - 1)$$

$\underbrace{\hspace{10em}}_{\text{Zero Mean.}}$

$$= w + (\beta + \gamma) r_{t-1}^2 - \beta \eta_{t-1} + \eta_t$$

(ARMA形式)

估计方法:

$$l_T(w, \beta, \gamma) = C - \frac{1}{2} \sum_{t=1}^T \frac{r_t^2}{6_t^2(w, \beta, \gamma)} - \frac{1}{2} \sum_{t=1}^T \log 6_t^2(w, \beta, \gamma)$$

使用数值方法求解

IGARCH: $\beta + \gamma = 1$, 冲击对波动的影响不会衰减, 方差不存在
(贴近现实情况)

$$\text{IGARCH}(1, 1): 6_t^2 = w + \beta 6_{t-1}^2 + (1 - \beta) r_{t-1}^2$$

① 6_t^2 是“鞅”, 即, $E(6_t^2 | \mathcal{F}_{t-1}) = 6_{t-1}^2$

$$\text{由于 } E(r_{t-1}^2 | \mathcal{F}_{t-1}) = 6_{t-1}^2$$

$$(E(r_{t-1}^2 | \mathcal{F}_{t-1}) = 6_{t-1}^2 E(\varepsilon^2) = 6_{t-1}^2)$$

$$\therefore E(6_t^2 | \mathcal{F}_{t-1}) = w + 6_{t-1}^2$$

② 方差不存在

③ 水平突变 $6_t^2 \approx 6_{t-1}^2 + \text{shock}$

GARCH不能识别 Leverage Effect, Volatility Feedback

GJR-GARCH:

$$6_t^2 = w + \beta 6_{t-1}^2 + \gamma r_{t-1}^2 + \delta r_{t-1}^2 \mathbb{1}(r_{t-1} < 0)$$

↓
只在负收益时放大波动

E-GARCH: 从“结构”上产生非对称性

$$\ln(6_t^2) = w + \lambda \ln(6_{t-1}^2) + g(\varepsilon_{t-1});$$

$$g(\varepsilon_t) = \theta \varepsilon_t + \gamma (|\varepsilon_t| - E|\varepsilon_t|)$$

↓
方向效应

↓
偏离水平

多变量 GARCH

$$x_t = \sigma_t \eta_t, \quad \sigma_t^2 = \alpha_0 + \sum_{i=1}^m \alpha_i x_{t-i}^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2$$

$$X_t = (x_{t1}, x_{t2}, \dots, x_{tN})'$$

$$X_t = \Sigma_t^{1/2} \cdot \eta_t$$

$$\text{CCC: } \Sigma_t = D_t R D_t; \quad D_t = \begin{pmatrix} \sigma_{1,t} & 0 \\ 0 & \sigma_{2,t} \end{pmatrix} \quad R = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$$

$$Q1: (a) \gamma(k) = \text{Cov}(x_t, x_{t-k}) = \text{Cov}((0.5)^k x_{t-k} + \sum_{i=0}^{k-1} \varepsilon_{t-i} (0.5)^i, x_{t-k})$$

$$= (0.5)^k \text{Var}(x_{t-k})$$

$$\text{Var}(x_t) = \frac{1}{4} \text{Var}(x_{t+1}) + 1, \therefore \text{Var}(x_t) = \frac{4}{3}, \gamma(k) = \frac{4}{3} (0.5)^k$$

$$(b) \gamma(k) = \text{Cov}(y_t, y_{t-k}) = \text{Cov}(x_t + \eta_t, x_{t-k} + \eta_{t-k})$$

$$k=0 \text{ 时, } \text{Cov}(x_t + \eta_t, x_t + \eta_t) = \text{Var}(x_t) + \text{Var}(\eta_t) = 1 + \frac{4}{3} = \frac{7}{3}$$

$$k \neq 0 \text{ 时, } = (0.5)^k \cdot \frac{4}{3}$$

$$(c) (1 - \varphi L) z_t = (H \theta L) u_t$$

$$z_t = \frac{H \theta L}{1 - \varphi L} u_t = (H \theta L) \cdot \sum_{i=0}^{\infty} (\varphi L)^i \cdot u_t$$

$$= (H \theta L) \cdot \sum_{i=0}^{\infty} \varphi^i \cdot u_{t-i}$$

$$= \sum_{i=0}^{\infty} \psi_i \cdot \varepsilon_{t-i}$$

$$\psi_0 = 1, \psi_1 = \theta + \varphi, \psi_2 = \theta\varphi + \varphi^2, \dots$$

$$\psi_j = \theta \cdot \varphi^{j-1} + \varphi^j, j \geq 1$$

$$\therefore \text{Var}(z_t) = \sigma^2 \cdot \sum_{i=0}^{\infty} (\psi_i)^2$$

$$= \sigma^2 \cdot \left(1 + \sum_{j=1}^{\infty} (\theta \varphi^{j-1} + \varphi^j)^2 \right)$$

$$= \sigma^2 \left(1 + \sum_{j=1}^{\infty} (\theta^2 \varphi^{2j-2} + 2 \varphi^{2j-1} \theta + \varphi^{2j}) \right)$$

$$= \sigma^2 \left(1 + \theta^2 \cdot \frac{1}{1 - \varphi^2} + 2\theta \cdot \frac{\varphi}{1 - \varphi^2} + \frac{\varphi^2}{1 - \varphi^2} \right)$$

$$= \sigma^2 \cdot \frac{\theta^2 + 2\theta\varphi + 1}{1 - \varphi^2}$$

$$\text{Cov}(z_t, z_{t-1}) = \text{Cov}\left(\sum_{i=0}^{\infty} \psi_i \varepsilon_{t-i}, \sum_{i=0}^{\infty} \psi_i \varepsilon_{t-1-i}\right)$$

$$= \sigma^2 \sum_{i=0}^{\infty} \psi_i \psi_{i+1}$$

$$= \sigma^2 \left(\theta + \varphi + \sum_{i=1}^{\infty} \psi_i \psi_{i+1} \right)$$

$$= \sigma^2 \left(\theta + \varphi + \sum_{i=1}^{\infty} (\theta \varphi^{i-1} + \varphi^i) (\theta \varphi^i + \varphi^{i+1}) \right)$$

$$= \sigma^2 \left(\theta + \varphi + \sum_{i=1}^{\infty} (\theta^2 \varphi^{2i-1} + \varphi^{2i+1} + 2\theta \varphi^{2i}) \right)$$

$$= \sigma^2 \left(\theta + \varphi + \frac{\theta^2 \varphi + \varphi^3 + 2\theta \varphi^2}{1 - \varphi^2} \right)$$

$$= \sigma^2 \cdot \frac{\theta + \varphi + \theta^2 \varphi + \theta \varphi^2}{1 - \varphi^2} = \frac{\sigma^2 (\theta + \varphi)(\theta \varphi + 1)}{1 - \varphi^2}$$

$$\begin{aligned}\gamma(h) &= \text{Cov}(z_t, z_{t-h}) = \text{Cov}(\varphi z_{t-1} + u_t + \theta u_{t-1}, z_{t-h}) \\ &= \varphi \cdot \text{Cov}(z_{t-1}, z_{t-h}) \\ &= \varphi \gamma(h-1)\end{aligned}$$

Q2.

Q3. (a) $E(u_t) = E(E(u_t|x)) = 0$

$$E(u_t x_t) = E(E(u_t x_t | x)) = E(x_t \cdot E(u_t | x)) = E(0) = 0$$

$$\text{Cov}(x_t, u_t) = E(x_t u_t) - E(x_t) E(u_t) = 0$$

(b) $\rho_j = \frac{\gamma_j}{\gamma_0} = \frac{\gamma_j}{\sigma^2}$

$j=1$ 时.

$$\text{Cov}(u_t, u_s) = E(\text{Cov}(u_t, u_s | x)) + \text{Cov}(E(u_t | x), E(u_s | x))$$

(d) $\hat{\beta}_1 = \frac{\rho_{xy}}{\rho_{xx}} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$

$$y_i - \bar{y} = \beta_1 (x_i - \bar{x}) + u_i$$

$$\therefore \hat{\beta}_1 = \frac{\sum \beta_1 (x_i - \bar{x})^2 + \sum (x_i - \bar{x}) u_i}{\sum (x_i - \bar{x})^2}$$

$$= \beta_1 + \frac{\sum (x_i - \bar{x}) u_i}{\sum (x_i - \bar{x})^2}$$

$$\therefore E(\hat{\beta}_1 | x) = \beta_1$$

(e) $\text{Var}(\hat{\beta}_1 | x) = \frac{1}{\sum (x_i - \bar{x})^2} \text{Var}(\sum (x_i - \bar{x}) u_i)$

$$\text{Var}(\sum z_i u_i) = \sum_{i=1}^T z_i^2 \cdot \sigma^2 + 2 \sum_{i=1}^T z_i z_{i+1} \frac{\sigma^2}{2}$$

Q4. (a) $E(y_t) = 0, \text{Var}(y_t) = \sigma^2 + \theta^2 \sigma^2 = (1 + \theta^2) \sigma^2$

$$\gamma(h) = \text{Cov}(y_t, y_{t-h}) = \text{Cov}(\varepsilon_t + \theta \varepsilon_{t-1}, \varepsilon_{t-h} + \theta \varepsilon_{t-h-1})$$

① 当 $t-1 > t-h$ 即 $h > 1$ 时, $\gamma(h) = 0$

② $h=1$. $\gamma(1) = \theta \sigma^2$

③ $h=0$. $\gamma(0) = (1 + \theta^2) \sigma^2$

$$P(h) = \begin{cases} 1, & h=0 \\ \frac{\theta}{1-\theta}, & h=1 \\ 0, & h>1 \end{cases}$$

$$\begin{aligned} (b) \quad \varepsilon_t &= y_t - \theta \varepsilon_{t-1} = y_t - \theta (y_{t-1} - \theta \varepsilon_{t-2}) \\ &= \dots \\ &= \sum_{k=0}^{\infty} \theta^k y_{t-k} \end{aligned}$$

$$\begin{aligned} (c) \quad \varepsilon_t &= y_t - \theta \varepsilon_{t-1} = y_t - \theta y_{t-1} + \theta^2 \varepsilon_{t-2} \\ &= \dots \\ &= \sum_{k=0}^{t-1} \theta^k y_{t-k} + \theta^t \varepsilon_0 \\ &= \sum_{k=0}^{t-1} \left(-\frac{1}{2}\right)^k y_{t-k} + \left(-\frac{1}{2}\right)^t \varepsilon_0 \end{aligned}$$

$$\begin{aligned} (d), \quad y_{t+1} &= \varepsilon_{t+1} + \theta \varepsilon_t \\ E(y_{t+1} | \varepsilon_0, y_1, y_2, \dots, y_t) &= \sum_{k=0}^T \theta^k y_{t-k} + \theta^{t+1} \varepsilon_0 + \theta \left(\sum_{k=0}^{T-1} \theta^k y_{t-k} + \theta^t \varepsilon_0 \right) \end{aligned}$$

$$PAM: C_t = -0.7 + 0.86y_t + 0.21C_{t-1} + \varepsilon_t$$

$$\text{等式: } C_t^* = \alpha + \beta y_t$$

$$C_t - C_{t-1} = \lambda (C_t^* - C_{t-1}) \quad \lambda: \text{调整速度} \quad \text{调整关系}$$

$$ARDL: Y_t = \alpha + \sum_{i=1}^p \phi_i Y_{t-i} + \sum_{j=0}^q \beta_j X_{t-j} + u_t \quad \text{动态滞后关系}$$

$$ECM: \Delta C_t = 0.97 \Delta y_t - 0.34 (C_{t-1} - 1.06 y_{t-1}) + u_t \quad \text{短期+长期均衡}$$

$$F = \frac{(SSR_R - SSR_U)/k}{SSR_U/(n-k_U)}$$

k : 约束个数

n : 样本量

k_U : 无约束变量数

1. (1). $(1-L)p_t = \varepsilon_t$, 特征方程 $1-\lambda=0$, $\lambda=1$

有单位根, 不平稳

(2) 不能, 模型有单位根, 过去的影响将永远存在于系统中,

无法消除

不能准确预测

(3) $\Delta p_t = \varepsilon_t \sim N(0, \sigma^2)$

Δp_t 是平稳的

GARCH: $\sigma_t^2 = \alpha + \beta \sigma_{t-1}^2 + \gamma r_{t-1}^2$, $r_{t-1} = \sigma_{t-1} \varepsilon_{t-1}$